
1. SOCIAL MEDIA ANALYTICS: AN OVERVIEW

The slide features a minimalist design with two thick teal horizontal lines at the top and bottom. Between these lines, there are two thin light blue horizontal lines. The title is centered in a bold, orange, sans-serif font. Below the title, there are two short, thick, olive-green horizontal dashes, one on the left and one on the right, positioned at the same vertical level.

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Social Media Analytics: Overview

- Social media analytics is the process of **extracting valuable insights** from **social media data to perform decision making**.
- By analyzing social media data, **businesses can increase brand loyalty, generate leads, drive traffic, and make forecasts**.
- Social media analytics can also be used to increase awareness of a brand and drive users to a website for the latest news and information.

Purpose of Social Media Analytics

- The purpose of social media analytics is to enable **informed decision making** by leveraging social media data. This involves answering questions such as:
 - What are customers saying about a brand or product on social media?
 - Which content posted on social media is resonating more with customers?
 - How can social media data be used to improve products or services?
 - Is the social media conversation about a company, product, or service positive, negative, or neutral?
 - How can social media be used to promote brand awareness?
 - Who are the influential followers, fans, and friends on social media?
 - Who are the influential nodes (people and organizations) on social media and their position in the network?
 - Which social media platforms are driving the most traffic to a corporate website?
 - Where are the social media customers located geographically?
 - Which keywords and terms are trending on social media?
 - How active is social media in a business and how many people are connected with the company?
 - Which websites are connected to a corporate website?
 - How are competitors performing on social media?

Social Media Vs Traditional Business Analytics

- The main difference between social media analytics and traditional business analytics is the **source, type, and nature of the data** being mined.
- **Social media analytics** involves the collection, analysis, and interpretation of **semistructured and unstructured social media data**, while **traditional business analytics** uses **structured and historical data**.
- Social media data is **diverse, high volume, real-time, and stored in third-party databases** in semistructured and unstructured formats, while traditional business data is mostly stored in **databases and spreadsheets in machine-readable** format.
- Social media data is **socialized in nature and originates from the public internet**, while traditional business data is **bureaucratic and formal in nature and is controlled by organizations**.
- **The value of** social media data is determined by the **extent to which it is shared with other social entities**, while the value of traditional business data is often **confined within organizational databases and serves as a source of competitive advantage**.

Social Media Vs Traditional Business Analytics

Social Media Analytics	Business Analytics
Semistructured and unstructured data	Structured data
Data is not analytical friendly	Data is analytical friendly
Real-time data	Mostly historical data
Public data	Private data
Stored in third-party databases	Stored in business-owned databases
Boundary-less data (i.e., Boundary within the Internet)	Bound within the business intranet

Social Media Vs Traditional Business Analytics

Social Media Analytics	Business Analytics
Data is high volume	Data is medium to high volume
Highly diverse data	Uniform data
Data is widely shared over the Internet	Data is only shared within organizations
More sharing creates greater value/impact	Less sharing creates more value
No business control over data	Tightly controlled by business
Socialized data	Bureaucratic data
Data is informal in nature	Data is formal in nature

Seven Layers of Social Media Analytics

- Social media has a **minimum has seven layers of data** (Figure 2).
- Each layer carries potentially valuable information and insights that can be harvested for business intelligence purposes.
- Out of the seven layers, some are visible or easily identifiable (e.g., text and actions) and other are invisible (e.g., social media and hyperlink networks).
- The following are seven social media layers
 - 1. Text
 - 2. Networks
 - 3. Actions
 - 4. Hyperlinks
 - 5. Mobile
 - 6. Location
 - 7. Search engines

Seven Layers of Social Media Analytics



Figure 2. Seven layers of social media analytics

Seven Layers of Social Media Analytics

Social media analytics involves analyzing data from seven layers:



Figure 2. Seven layers of social media analytics

1. **Text:** This includes the content of social media posts, such as **comments, tweets, blog posts, and Facebook status updates**. It is used to understand user sentiments and identify emerging themes and topics.
2. **Networks:** This includes the **connections between users** and the relationships between them, such as followers and friends on social media. It is used to identify influential nodes (people and organizations) and their position in the network.
3. **Actions:** This includes the **actions taken by users on social media, such as likes, comments, shares, and other interactions**. It is used to measure popularity, influence, and prediction in social media.

Seven Layers of Social Media Analytics

Social media analytics involves analyzing data from seven layers:

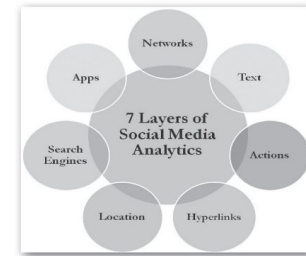


Figure 2. Seven layers of social media analytics

4. **Mobile:** This includes data related to the **use of social media on mobile devices, such as the type of device used, the operating system, and the location of the user.** It is used to measure and optimize user engagement with mobile applications.

5. **Hyperlinks:** This includes the **links between social media posts and other websites,** as well as the content of those linked websites. It is used to reveal Internet traffic patterns and sources of incoming or outgoing traffic.

6. **Location:** This includes data on the **geographical location of users** and the **location-specific content** they engage with on social media. It is used to mine and map the locations of social media users, content, and data.

7. **Search engines:** This includes data on **how users find and access social media content** through **search engines,** as well as the keywords they use in their searches. It is used to analyze historical search data for trends analysis, keyword monitoring, and advertisement history.

Social Media Analytics Types

There are three main types of social media analytics:

1. **Descriptive analytics:** This involves gathering and describing social media data in the form of **reports, visualizations, and clustering to understand a business problem**. Examples include actions analytics (e.g., number of likes, tweets, and views) and text analytics.
2. **Diagnostic analytics:** addresses the next logical question, “Why did this happen?”
3. **Predictive analytics:** This involves analyzing large amounts of accumulated social media data to **predict a future event**. For example, analyzing social media posts to predict future purchasing behavior or using historical website visits to predict future sales figures.
4. **Prescriptive analytics:** This involves suggesting the best action to take when handling a scenario based on analyzing patterns of behavior. It has not yet been widely applied to social media data.

Social Media Analytics Lifecycle

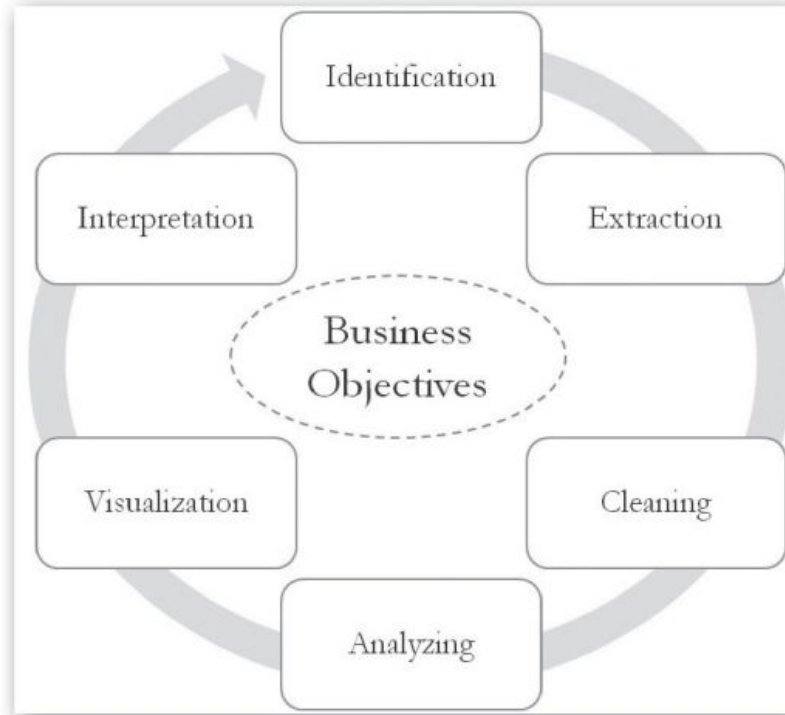


Figure 3. Social media analytics cycle

Social Media Analytics Lifecycle

The social media analytics process involves six steps to mine desired business insights from social media data. The process begins with defining business goals and objectives, and continues until these objectives are fully satisfied.

The six general steps are:

1. **Data collection:** Gathering relevant data from social media platforms.
2. **Data processing:** Cleaning and preparing the data for analysis.
3. **Data analysis:** Analyzing the data using various methods and tools to uncover insights.
4. **Insights generation:** Interpreting the results of the analysis to generate insights.
5. **Insights communication:** Communicating the insights to relevant stakeholders.
6. **Insights implementation:** Using the insights to inform business decisions and actions.

Step1 - Identification

- The identification stage of social media analytics involves **finding the right sources of data to analyze** in order to gain valuable business insights.
- The **data** should be aligned with the **business's objectives** and can come from both official business-owned platforms, such as social media accounts and blogs, and nonofficial platforms such as Google search trends or Twitter search stream data.
- It is important to consider the business objectives when identifying the sources and types of data to be analyzed.

Step2 - Extraction

- The extraction stage of social media analytics involves using **appropriate methods and tools to gather data from identified sources**. This can include manual data collection for small-scale data and automated extraction using APIs (application programming interfaces) for larger data sets.
- It is important to **consider privacy and ethical issues** when mining data from social media platforms and to have a clear social media privacy policy in place to ensure that data handling and extraction practices do not violate user privacy.
- Specialized tools may be needed to extract certain types of data, such as social network and hyperlink network data.

Step3 - Cleaning

- The cleaning step in social media analytics involves **removing unwanted data** from the collected data set.
- This can involve processes such as **coding, filtering, clustering, and natural language processing** to remove irrelevant data.
- Both **automated and manual techniques** may be used for cleaning, depending on the type of data and the desired level of accuracy.

Step 4- Analyzing

- The analyzing stage of social media analytics involves using clean data to **identify valuable insights for the business.**
- The approach and techniques used will depend on the type of data being analyzed and the tools and algorithms employed.
- It is important **to maintain the integrity of the data** while **extracting meaningful insights** and to have a good understanding of the capabilities of the tools being used.

Step 5 - Visualization

- The visualization step in social media analytics involves **creating visual representations of the results of the analysis**
- Visualization can help reveal hidden patterns, relationships, and trends in complex and large data sets
- Different types of data will result in different types of visualizations, including **network data, topical data, temporal data, and geospatial data**
- Other types of visualizations include **trees, hierarchical graphs, and heat maps**
- **Effective visualization** is important for effectively communicating the results of the analysis to top management

Step 6 : Interpretation

- Interpret and translate analytics results into a meaningful business problem.
- Two strategies or approaches used are:
 - 1) Producing easily consumable analytical results and
 - 2) Improving analytics consumption capabilities

Challenges to Social Network Analysis

- **Volume and Velocity** is a challenge
- **Diversity Challenge**- Social media users and content they generate are extremely diverse, multilingual, and vary across time and space.
- **Unstructuredness** as a challenge

Social Media Analytics Tools

Layer of social media	Example of tools
Text	Discovertext Lexalytics Demo :Video Link Archivist Twitonomy Netlytic LIWC Voyant
Actions	Lithium Twitonomy Google Analytics Video Demo : Video Link SocialMediaMineR
Network	NodeXL UCINET Pajek Netminer Flocker Netlytic Reach Mentionmapp

Mobile	Countly
	Mixpanel
	Google Mobile Analytics
	Google Fusion Table
Location	Tweepstmap
	Trendsmat
	Followerwonk
	Esri Maps Agos
Hyperlinks	Webometrics Analyst VOSON
Research Engines	Google Trends

Questions

- Why it is important for business managers to understand and mine social media data?
- What is social media analytics, and how it is different from traditional business analytics?
- Briefly explain the seven layers of social media data. Support your answer with examples.
- Explain the social media analytics cycle.
- What ethical issues should be considered when mining social media data?
- What are some main challenges to social media analytics?
- Compare different social media analytics tools available in the market and explain their strengths and weakness.